PRE-ALGEBRA (8TH) · UNIT 6: CONGRUENCE AND SIMILARITY

Tile Pattern Studio

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students design a lobby floor from one right-triangle tile, moving it with transformations, enlarging it with dilations, and using angle facts so the pattern fits.

Congruence, similarity, and angle relationships come together in a design students can draw and present.

Standards Addressed

- 8.G.A.1–3 Properties and coordinates of transformations
- 8.G.A.4 Similarity through dilations
- 8.G.A.5 Angle sum, exterior angles, parallel lines, AA similarity

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with tile floor photos. Part 1: rigid motions on the grid.
Day 2	Part 2: dilations and similarity, indirect measurement.
Day 3	Part 3: angle relationships and the final tile design.

Materials

- Student handout
- Grid paper
- Patty paper or tracing paper
- Colored pencils
- Optional: pattern blocks

How the Three Versions Differ

Level	What changes
CP2	Single transformations with rules given, dilation by 2, and basic angle facts.
CP1	Sequences of moves, area under dilation, indirect measurement, and angle equations.
Honors	Equivalent transformations, dilation about any center, mirror measurement, and tessellation proofs.

Watch For (Common Misconceptions)

- Switching x and y incorrectly in rotation rules.
- Thinking a dilated figure is congruent.
- Adding instead of matching corresponding angles.

Extension Ideas

- Create the design in a geometry app and animate the moves.
- Find real tile patterns at school and identify the transformations.
- Explore which irregular pentagons can tile a floor.

Answer Key – CP2

Part 1: Slide and Flip

#	Task	Answer
1	Translate the tile 5 units right and 2 units up. Give the new coordinates.	A′(6, 3), B′(9, 3), C′(6, 5).
2	Reflect the original tile over the y-axis. Give the new coordinates.	A′(−1, 1), B′(−4, 1), C′(−1, 3).
3	Are the new tiles congruent to the original? Explain.	Yes. Translations and reflections are rigid motions, so lengths and angles stay the same.

Part 2: Turn and Grow

#	Task	Answer
4	Rotate the tile 90° counterclockwise about the origin. Use (x, y) → (−y, x).	A′(−1, 1), B′(−1, 4), C′(−3, 1).
5	Dilate the tile by a scale factor of 2 from the origin.	A′(2, 2), B′(8, 2), C′(2, 6).
6	Side AB is 3 units long. How long is A′B′ after the dilation? Is the new tile congruent or similar?	A′B′ = 6 units. The tile is similar (same shape, different size), not congruent.

Part 3: Angle Facts

#	Task	Answer
7	The tile has a 90° angle and a 34° angle. Find the third angle.	180° − 90° − 34° = 56°.
8	Grout lines form parallel lines cut by a transversal. One angle is 115°. Find a corresponding angle and a same-side interior angle.	Corresponding angle: 115°. Same-side interior angle: 180° − 115° = 65°.
9	Design a pattern using at least two different transformations of the tile. Describe each move.	Answers vary; each move should be named with its rule (for example, translate 3 right, then reflect over the x-axis).

Reflection

#	Prompt	Look for
R1	Which transformations make a congruent figure? Which make a similar figure?	Translations, reflections, and rotations make congruent figures; dilations make similar figures.
R2	How do parallel lines help you find angles in the pattern?	Corresponding and alternate angles are equal, and same-side interior angles add to 180°.

Answer Key – CP1

Part 1: Sequences of Moves

#	Task	Answer
1	Reflect the tile over the x-axis, then translate it 3 units left. Give the final coordinates.	After reflecting: (1, −1), (4, −1), (1, −3). After translating: A″ (−2, −1), B″(1, −1), C″(−2, −3).
2	Write the rule for a translation 5 right and 2 up.	(x, y) → (x + 5, y + 2).
3	Rotate the tile 180° about the origin.	(x, y) → (−x, −y): A′(−1, −1), B′(−4, −1), C′(−1, −3).

Part 2: Similar Tiles

#	Task	Answer
4	Dilate the tile by 1.5 from the origin. Find the new coordinates and compare the areas.	A′(1.5, 1.5), B′(6, 1.5), C′(1.5, 4.5). Area: $\frac{1}{2}(3)(2) = 3 \rightarrow \frac{1}{2}(4.5)(3) = 6.75$ ($1.5^2 = 2.25$ times).
5	Are triangles with sides 3, 4, 5 and 4.5, 6, 7.5 similar? Explain.	Yes: each side is multiplied by 1.5.
6	A 6-foot column casts a 4-foot shadow while a flagpole casts a 14-foot shadow. How tall is the flagpole?	$\frac{6}{4} = \frac{h}{14} \rightarrow$ h = 21 feet.

Part 3: Solving with Angles

#	Task	Answer
7	A triangular tile has angles 47° and 68°. Find the third angle and the exterior angle next to it.	Third angle 65°; exterior angle 180° − 65° = 115° (also 47° + 68°).
8	Alternate interior angles measure (3x + 10)° and (5x − 30)°. Find x and the angle.	3x + 10 = 5x − 30 → x = 20; the angle is 70°.
9	Explain why two tiles with angles of 40° and 75° must be similar.	Their third angles are both 65°, so all angles match (AA similarity).

Reflection

#	Prompt	Look for
R1	Why does the area grow by the square of the scale factor?	Both the base and the height are multiplied by the scale factor, so the area is multiplied by its square.
R2	Why is AA enough to show triangles are similar?	If two angles match, the third must match too (angle sum 180°), so the triangles have the same shape.

Answer Key – Honors

Part 1: Equivalent Moves

#	Task	Answer
1	Show that reflecting over the y-axis and then over the x-axis is the same as one rotation. Which one?	(x, y) → (−x, y) → (−x, −y), the rule for a 180° rotation about the origin.
2	Find a transformation (or sequence) that maps ABC to A″(−1, −1), B″(−1, −4), C″(−3, −1).	Reflection over the line y = −x: (x, y) → (−y, −x). (Also: rotate 90° counterclockwise, then reflect over the x-axis.)
3	Dilate the tile by a scale factor of 2 centered at C(1, 3).	Image = C + 2(P − C): A′(1, −1), B′(7, −1), C′(1, 3).

Part 2: Similarity at Work

#	Task	Answer
4	A tile is dilated by a scale factor of 3. The original area is 3 square units. Find the new area.	$3^2 \times 3 = 27$ square units.
5	A student's eyes are 5 feet high. She stands 4 feet from a floor mirror and sees the top of a building 48 feet from the mirror. How tall is the building?	$\frac{5}{4} = \frac{h}{48}$ → h = 60 feet.
6	Two grout lines are parallel and two transversals meet at a point, forming two triangles. Explain why the triangles are similar.	The vertical angles are equal and alternate interior angles are equal, so two pairs of angles match (AA).

Part 3: Which Shapes Tile?

#	Task	Answer
7	Explain why regular hexagons tile a floor but regular pentagons do not.	Hexagon interior angle 120°: 3 × 120° = 360°. Pentagon angle 108°: 360 ÷ 108 is not a whole number.
8	A triangle has angles (2x)°, (3x + 5)°, and (4x − 14)°. Find each angle.	9x − 9 = 180 → x = 21: 42°, 68°, 70°.
9	Design a tessellation that uses the original tile and its images under two different transformations. Describe the moves and show why the angles fit.	Answers vary; angles meeting at each vertex must add to 360°.

Reflection

#	Prompt	Look for
R1	How can you prove two sequences of transformations are equivalent?	Write each as a coordinate rule and show the rules give the same image for every point.
R2	Why do any triangle and any quadrilateral tile a floor?	Six copies of a triangle fit around a point (2 × 180° = 360°) and four copies of a quadrilateral use its 360° angle sum.